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Analog Electronics (RAPC-209)

Test-I (MM: 20)

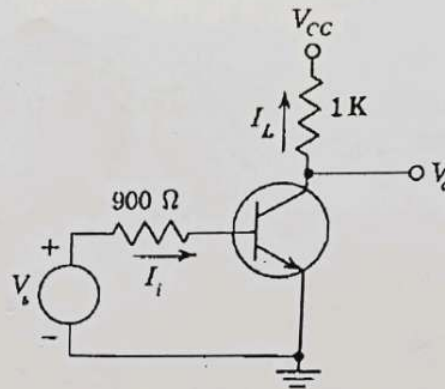
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1. The hybrid- π parameters of the transistor used in the circuit shown below are given as

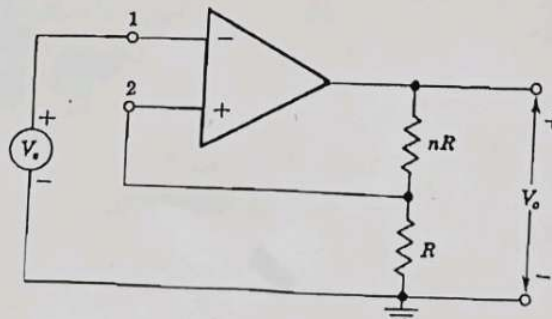
Hybrid- π Parameter Values Typical magnitudes for the elements of the hybrid- π model at room temperature and for $I_C = 1.3$ mA are

$$\begin{array}{llll} g_m = 50 \text{ mA/V} & r_{bb'} = 100 \Omega & r_{b'e} = 1 \text{ K} & r_{b'c} = 4 \text{ M} \\ r_{ce} = 80 \text{ K} & C_c = 3 \text{ pF} & C_\pi = 100 \text{ pF} & \end{array}$$

Using Miller's theorem and the approximate analysis, compute the upper 3-dB frequency of the current gain $A_i = I_L / I_i$



2. Prove that over the range of frequencies from $10f_L$ to $0.1f_H$ the voltage amplification is constant to within 0.5% and the phase shift to within ± 0.1 rad. Assume the standard low and high frequency responses as that from a simple RC circuits.
3. Draw the h -parameter model of the common mode gain in a Differential amplifier. Without explicitly solving for A_c show that it must be zero if $\frac{h_{fe}}{h_{oe}} = 2R_e$
4. For the base differential-input amplifier shown in the Figure below, assume infinite input resistance, zero output resistance, and finite differential gain $A_v = V_o / (V_1 - V_2)$. i) obtain an expression for the gain $A_{vf} = V_o / V_s$ ii) show that $\lim A_{vf} = n + 1, A_v \rightarrow \infty$



OR

The Schmitt trigger shown in the Figure is modified to include two clamping Zener diode with Zener voltage V_Z across the output. If $V_Z = 4\text{ V}$ and $A_v = 5000$ and if the threshold levels desired are $6 \pm 0.5\text{ V}$, find i) R_2/R_1 ii) the loop gain iii) V_R

